



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-2027
COURSE HANDOUT – Part-II

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : CE G545
Course Title : AIRPORT PLANNING & DESIGN
Instructor-in-Charge : G MUTHUKUMAR

1. Course Description: The course deals with complete planning and design of airports with greater emphasis on planning not restricted to airports, but air transport at large. The course will also touch base on several other connected aspects of air transportation such as forecasting air transport demand, evolution of aircraft and pavements, air transportation business, sustainability aspects etc.,

2. Scope and Objective of the Course: The scope of the course is to know the various aspects of airport planning and design. This course is expected to provide a broad overview on various topics as mentioned below.

- To understand briefly air transport and aviation industry [Module I]
- To understand the need to forecast air transport demand [Module II]
- To get the overall idea about aircraft and its influence on entire planning [Module III]
- To get a comprehensive overview about airport system and master planning [Module IV]
- To get an overview about the factors affecting the airport capacity [Module V]
- To comprehensively analyze the airside configuration and geometric design [Module VI]
- To understand the structural pavement design and airport drainage [Module VII]
- To get a quick overview about different visual aids in airport and in airspace [Module VIII]
- To acknowledge the importance of maintenance and rehabilitation of pavement [Module IX]
- To know the parameters influencing the airport performance including benchmarking [Module X]

3. Course learning outcomes:

- Complete comprehensive understanding of airport planning process
- Structural pavements, maintenance and failures
- Detailed insights into various guidelines and standards

4. Text Books & Reference Books:

TB1. Planning and Design of Airports by Robert Horonjeff et al Fifth Edition, McGraw Hill

TB2. Khanna, S.K., Arora M.G., and Jain, S.S. (1999) “Airport Planning and Design, Nemchand.



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RB1: Airport Engineering by Norman J Ashford et al. 4th Edition, Wiley Publications

- RB2: Airport Engineering (Planning and Design) by Subhash C. Saxena, CBS Publishers & Distributors Pvt Ltd.
- RB3: Rangwala (2016) “Airport Engineering” Charotar, 16th edition
- RB4: Subramanian, K.P. (2010) “Highway Railway Airport and Harbour Engineering” Scitech Publications (India) Pvt, Ltd, Chennai.

5. Method of conducting the course:

- Both blackboard and PPT will be used during the lecture session
- Lecture notes will be uploaded on Nalanda for reference

6. Course Plan:

Week No.	Module No.	Lecture No.	Lecture Session	Reference	Learning outcomes
1	1	1-3	• Need for air transport • Advantages and disadvantages of air transport • Types of air transport • Governmental and Non-governmental organizations related to aviation industry • Functions of AAI • Revenues and expenditures at Airports	T1,T2	Introduction to air transportation and aviation business
2	2	4-7	• Factors related to air transport demand • Conventional airport forecast methods • Demand forecast framework and models • Overview of components of airside and landside infrastructure	T1,T2	Air Transport Demand forecasting
3-4	3	8-12	• Characteristics of aircraft • Influence of aircraft on airports and runway length • Other airport layout factors • Airport capacity factors	T1,T2	Aircrafts and Airports
5	4	13-16	• Aviation system planning • Airport system planning • Hierarchy of planning • Airport Layout Design • Airport Site Selection	T1,T2	Airport System Planning & Airport Master Planning
6-7	5	17-22	• Level of service • Airside capacity and runway capacity • Taxiway capacity • Gate capacity • Capacity-Delay analysis • Passenger Terminal – Functions, Terminal Design	T1,T2	Airport Capacity
8-9	6	23-27	• Airfield and Runway configuration and orientation	T1,T2	Airside configuration



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

			- Runway Length - Obstructions to airspace - Clearways and stop ways - Longitudinal grade design for runways, taxiways and stop ways - Taxiway Design, Holding Aprons and Terminal Aprons		and Geometric Design of Airside
10-11	7	28-33	- Surface and Sub-surface drainage - Surface Hydrology - Pavement Surface drainage - Flexible pavement design methods (conventional) - Rigid pavement design methods (conventional) - Introduction to FAARFIELD design process	T1,T2	Airport Drainage and Structural Pavement Design
12	8	33-36	- Basics of airport markings, runway markings, taxiway markings - Lighting fixtures, light intensity and control, emergency lighting - Type of signs - Taxiway designation and sign system - Purpose of air traffic control - Airport traffic control tower - Airways and area navigation	T1,T2	Airport Visual Aids & Air Traffic control
13	9	37-40	- Defects in airport pavements - Elements of pavement evaluation - Maintenance Management System - Reporting of pavement strength - Airport Pavement overlays	T1,T2	Maintenance, Evaluation and strengthening of airport pavements
14	10	41-42	- Measuring airport economic performance - Internal benchmarking - External benchmarking - Safety, Quality of service, Cost effectiveness	T1,T2	Airport performance assessment

7. Evaluation Scheme:

Component	Weightage (%)	Duration	Date & Time	Comments
Announced Class Quiz 1	5%	50 minutes (each)	August 24 th , 2026 (Class hour)	Open book
Announced Class Quiz 2	5%	50 minutes (each)	September 23 rd , 2026 (Class hour)	Open book
Announced Class Quiz 3	5%	50 minutes (each)	November 16 th , 2026 (Class hour)	Open book
Course Project	25%	1 month	November 15 th , 2026	Open book
Mid-Semester Examination	25%	1.5 hours	<TEST_1>	Open Book



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
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Comprehensive Examination	35%	3 hours	<TEST_C>	Closed book/Open Book
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Criterion for NC: If the candidate misses one particular component completely, NC will be awarded. Also, if the candidate is appearing for the sake of appearing, NC will be awarded.

8. Chamber Consultation Hour: Thursday (6 PM – 7 PM).

9. Make-up Policy: Make-up will be granted only to genuine cases. For cases related to illness, proper documentary evidence is essential.

10. Note (if any): Students are expected to attend the classes sincerely. Notice, if any, concerning the course will be displayed on the Nalanda LMS.

Instructor-in-charge
CE G545